10.6 (a) For incident wave with polarization $(\vec{\epsilon}_1 \pm i\vec{\epsilon}_2)/\sqrt{2}$, the scattered electric field is, by Eq. (10.57),

$$\vec{E}_{sc} = \sqrt{\frac{\pi}{2}} \sum_{l=1}^{\infty} i^l \sqrt{2l+1} \left[\alpha_\pm(l) h_l^{(1)}(kr) \vec{X}_{l,\pm 1} \pm \frac{\beta_\pm(l)}{k} \nabla \times h_l^{(1)}(kr) \vec{X}_{l,\pm 1} \right].$$

Notice the difference of a $1/\sqrt{2}$ factor. Using Eq. (10.60), the second term becomes

$$\nabla \times h_l^{(1)}(kr) \vec{X}_{l,\pm 1} = \frac{i \vec{n} \sqrt{l(l+1)}}{r} h_l^{(1)}(kr) Y_{l,\pm 1} + \frac{1}{r} \frac{\partial}{\partial r} \left[r h_l^{(1)}(kr) \right] \vec{n} \times \vec{X}_{l,\pm 1}$$

Now, using the asymptotic expression for $h_l^{(1)}(kr) = (-i)^{l+1} \frac{e^{ikr}}{kr}$, the scattered field is

$$\vec{E}_{sc} = \frac{e^{ikr}}{r} \frac{1}{ik} \sqrt{\frac{\pi}{2}} \sum_{l=1}^{\infty} \sqrt{2l+1} \left[\alpha_\pm(l) \vec{X}_{l,\pm 1} \pm \frac{i\beta_\pm(l)}{k} \frac{\sqrt{l(l+1)}}{r} Y_{l,\pm 1} \vec{n} \pm i\beta_\pm(l) \vec{n} \times \vec{X}_{l,\pm 1} \right].$$

By the definition of $\vec{F}(\vec{k},\vec{k}_0)$ and also notice that $E_0 = 1$, we will obtain

$$\vec{f} = \frac{1}{ik} \sqrt{\frac{\pi}{2}} \sum_{l=1}^{\infty} \sqrt{2l+1} \left[\alpha_\pm(l) \vec{X}_{l,\pm 1} \pm i\beta_\pm(l) \vec{n} \times \vec{X}_{l,\pm 1} \right].$$

(b) By the optical theorem,

$$\sigma_t = \frac{4\pi}{k} \operatorname{Im}\left[\vec{\epsilon}_0^{\,*} \cdot \vec{f}(\vec{k} = \vec{k}_0) \right].$$

Here, $\vec{\epsilon}_0 = (\vec{\epsilon}_1 \pm i\vec{\epsilon}_2)/\sqrt{2}$. From solution to Prob. 10.2, we know

$$\vec{X}_{lm} = \frac{1}{\sqrt{l(l+1)}} \begin{pmatrix} \frac{1}{2}\left(\sqrt{(l-m)(l+m+1)}\, Y_{l,m+1} + \sqrt{(l+m)(l-m+1)}\, Y_{l,m-1} \right) \\ \frac{1}{2i}\left(\sqrt{(l-m)(l+m+1)}\, Y_{l,m+1} - \sqrt{(l+m)(l-m+1)}\, Y_{l,m+1} \right) \\ m Y_{lm} \end{pmatrix}$$

For $m = +1$,

$$\vec{X}_{l,+1} = \frac{1}{\sqrt{l(l+1)}} \begin{pmatrix} \frac{1}{2}\left(\sqrt{(l-1)(l+2)}\, Y_{l,2} + \sqrt{l(l+1)}\, Y_{l,0} \right) \\ \frac{1}{2i}\left(\sqrt{(l-1)(l+2)}\, Y_{l,2} - \sqrt{l(l+1)}\, Y_{l,0} \right) \\ Y_{l,1} \end{pmatrix},$$

$m = -1$

$$X_{l,-1} = \frac{1}{\sqrt{l(l+1)}} \begin{pmatrix} \frac{1}{2}\left(\sqrt{l(l+1)}\, Y_{l,0} + \sqrt{(l-1)(l+2)}\, Y_{l,-2} \right) \\ \frac{1}{2i}\left(\sqrt{l(l+1)}\, Y_{l,0} - \sqrt{(l-1)(l+2)}\, Y_{l,-2} \right) \\ -Y_{l,-1} \end{pmatrix}$$

For $\vec{k}=\vec{k}_0$, $\theta=\varphi=0$. Using the special value of the spherical harmonics,

$$Y_{lm}(0,\varphi) = \sqrt{\frac{2l+1}{4\pi}}\,\delta_{m,0}\,,$$ we have

$$\vec{X}_{l,+1}(0,0) = \frac{\sqrt{2l+1}}{4\sqrt{\pi}}(1,i,0)^T, \quad \vec{X}_{l,-1}(0,0) = \frac{\sqrt{2l+1}}{4\sqrt{\pi}}(1,-i,0)^T.$$

Also, $\vec{n}=\vec{\varepsilon}_3=(0,0,1)$, then

$$\vec{n}\times\vec{X}_{l,+1}(0,0) = \frac{\sqrt{2l+1}}{4\sqrt{\pi}}(-i,1,0)^T, \quad \vec{n}\times\vec{X}_{l,-1} = \frac{\sqrt{2l+1}}{4\sqrt{\pi}}(i,1,0)^T,$$

$$\vec{\varepsilon}_+^{\,*}\cdot\vec{X}_{l,+1} = \frac{\sqrt{2l+1}}{4}\sqrt{\frac{2}{\pi}}, \quad \vec{\varepsilon}_+^{\,*}\cdot(\vec{n}\times\vec{X}_{l,+1}) = -i\frac{\sqrt{2l+1}}{4}\sqrt{\frac{2}{\pi}}.$$

$$\vec{\varepsilon}_-^{\,*}\cdot\vec{X}_{l,-1} = \frac{\sqrt{2l+1}}{4}\sqrt{\frac{2}{\pi}}, \quad \vec{\varepsilon}_-^{\,*}\cdot(\vec{n}\times\vec{X}_{l,-1}) = i\sqrt{\frac{2l+1}{4}}\sqrt{\frac{2}{\pi}}.$$

Therefore,

$$\sigma_t = \frac{4\pi}{k}\cdot\frac{1}{k}\sqrt{\frac{\pi}{2}}\sum_{l=1}^{\infty}\operatorname{Im}\left[\frac{1}{i}\,\vec{\varepsilon}_\pm^{\,*}\cdot\left(\alpha_\pm(l)\,\vec{X}_{l,\pm1} \pm i\beta_\pm(l)\,\vec{n}\times\vec{X}_{l,\pm1}\right)\right]\sqrt{2l+1}$$

$$= \frac{4\pi}{k^2}\sqrt{\frac{\pi}{2}}\sum_{l=1}^{\infty}\sqrt{2l+1}\cdot\frac{\sqrt{2l+1}}{4}\cdot\sqrt{\frac{2}{\pi}}\operatorname{Im}\left[\frac{1}{i}\left(\alpha_\pm(l)+\beta_\pm(l)\right)\right]$$

$$= -\frac{\pi}{k^2}\sum_{l=1}^{\infty}(2l+1)\operatorname{Re}\left[\alpha_\pm(l)+\beta_\pm(l)\right]$$